



MATHEMATICS METHODS : UNITS 1 & 2, 2023
Test 3 – Counting & Probability, Exponential Functions and Sequences (10%)

PM

Time Allowed 27 minutes	First Name	Surname	Marks
	SOLUTIONS		27 marks

Circle your Teacher's Name: Mrs Alvaro Mr Bartoshefski Mr Buckland Mr Liu
 Mr Lockyer Mr Luzuk Mr Mohammadi
 Ms Narendranathan Mr Ng
 Mr Stephens Mr Tanday

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART A – CALCULATOR FREE

Question 1. [2 marks]

Write the value of xy in scientific notation when $x=2.5 \times 10^3$ and $y=5 \times 10^{-7}$

$$2.5 \times 10^3 \times 5 \times 10^{-7} = 12.5 \times 10^{-4} = 1.25 \times 10^{-3}$$

Question 2. [5 marks: 2, 3]

(a) Evaluate $\sqrt{4^{-5}}$

(b) Simplify and leave answers with positive indices

$$\frac{3^{4n} \times 18^{n+1}}{6^{3n-2}}$$

$$\sqrt{4^{-5}} = (\sqrt{4})^{-5} = (2)^{-5} = 1/2^5 = 1/32$$

$$\begin{aligned} \frac{3^{4n} \times (2 \times 3^2)^{n+1}}{(2 \times 3)^{3n-2}} &= 3^{4n+2n+2-3n+2} \times 2^{n+1-3n+2} \\ &= 3^{3n+4} \times 2^{-2n+3} = \frac{3^{3n+4} \times 2^3}{2^{2n}} = \frac{3^{3n+4}}{2^{2n-3}} \end{aligned}$$

Question 3. [6 marks: 3, 3]

Solve for x in each of the following, showing all working:

(a) $25^{2x+1} = \sqrt{125}$

$$\begin{aligned}5^{2(2x+1)} &= 5^{\frac{3}{2}} \\4x + 2 &= \frac{3}{2} \\x &= -\frac{1}{8}\end{aligned}$$

(b) $3^{2x+1} = 8(3^x) + 3$

$$\begin{aligned}3(3^x)(3^x) - 8(3^x) - 3 &= 0 \\(3 \times 3^x + 1)(3^x - 3) &= 0 \\3^x &= -\frac{1}{3} \text{ (no solution) or } 3^x = 3 \\X &= 1,\end{aligned}$$

Question 4. [4 marks: 2, 2]

The first three terms of an arithmetic progression are x , $5x - 8$ and $3x + 8$ respectively.

(a) Show that $x = 4$

$$\begin{aligned}(5x - 8) - (x) &= (3x + 8) - (5x - 8) \\ \Rightarrow 4x - 8 &= -2x + 16 \\ \Rightarrow 6x &= 24 \Rightarrow x = 4\end{aligned}$$

(b) Find the value of 21st term in this sequence?

$$a = 4, d = 8$$

$$21^{\text{th}} \text{ term} = 4 + 20 \times 8 = 164$$

Question 5. [5 marks: 1, 2, 2]

The binomial coefficients in the eighth row of Pascal's triangle are 1,8,28,56,... and so on.

(a) Deduce the value of

i) $\binom{8}{6}$

Using symmetry property $\binom{a}{b} = \binom{a}{a-b}$ then $\binom{8}{6} = \binom{8}{2} = 28$

ii) $\binom{9}{7}$

$$\binom{9}{7} = \binom{8}{6} + \binom{8}{7} = 28 + 8 = 36$$

$$\text{OR } \binom{9}{7} = 9!/7!2! = 36$$

$$\text{OR } \binom{9}{2} = 36$$

(b) The sum of all but one of the binomial coefficients in the eighth row of Pascal's triangle is 186. Determine, with justification, the value of the missing coefficient.

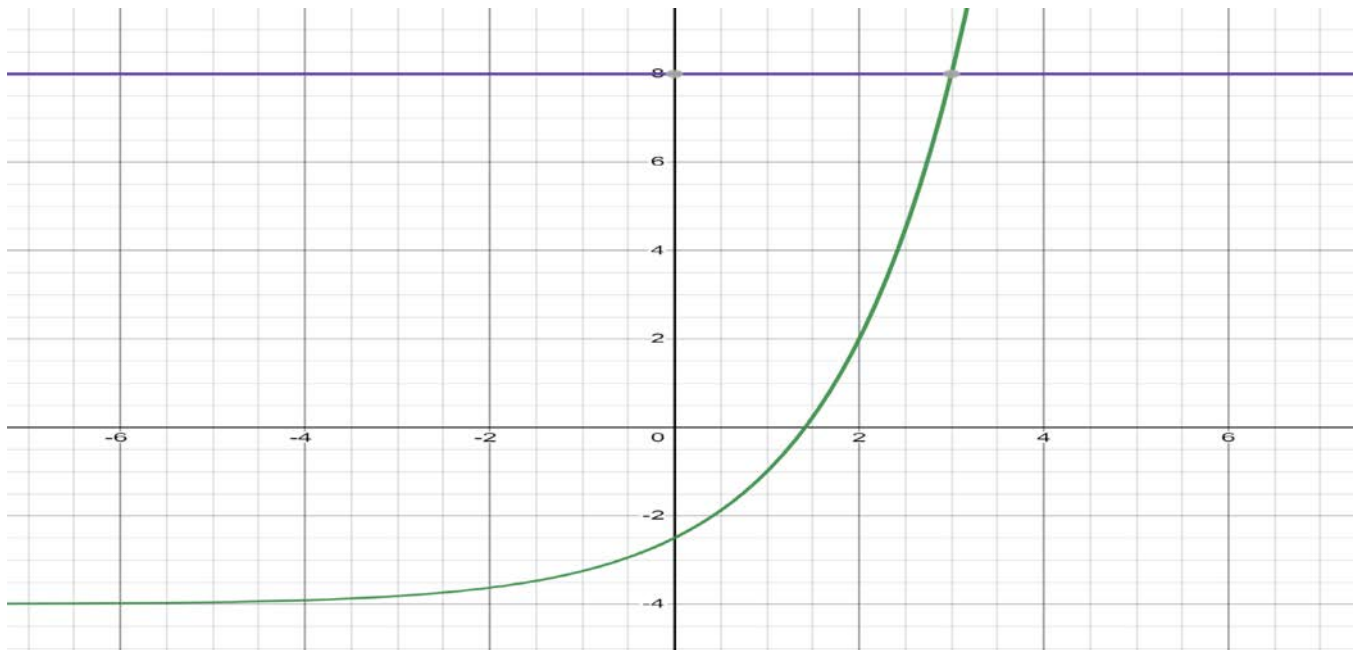
Sum of the row is $2^8 = 256$

Missing Coefficient is $256 - 186 = 70$

Question 6. [5 marks: 3, 2]

Consider, $y = 3(2^{x-1}) - 4$

(a) On the axes provided below, sketch this curve.



(b) Find the point of intersection between this curve and the line $y = 8$ ***algebraically***.

$$\begin{aligned} 3(2^{x-1}) - 4 &= 8 \\ X &= 3 \quad \text{POI}=(3,8) \end{aligned}$$



MATHEMATICS METHODS : UNITS 1 & 2, 2023
Test 3 – Counting & Probability, Exponential Functions and Sequences (10%)

PM

Time Allowed 23 minutes	First Name	Surname	Marks
	SOLUTIONS		23 marks

Circle your Teacher's Name: Mrs Alvaro Mr Bartoshefski Mr Buckland Mr Liu
 Mr Lockyer Mr Luzuk Mr Mohammadi
 Ms Narendranathan Mr Ng
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Assessment Conditions: *(N.B. Sufficient working out must be shown to gain full marks)*

- ❖ Calculators: Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART B – CALCULATOR ASSUMED

Question 7. [3 marks: 1, 2]

A chess club has 12 members, of which 5 are beginners, 3 are intermediate and the rest are advanced. The club has to select a group of 4 members at random to assist with a regional tournament.

- (a) Determine the number of different groups that can be selected.

$$\binom{12}{4} = 495$$

- (b) Determine the number of different groups that can be selected which contain at least 3 beginners.

Choose 3 beginners : $\binom{5}{3} \binom{7}{1} = 70$
Choose 4 beginners : $\binom{5}{4} \binom{7}{0} = 5$
Total $70 + 5 = 75$

Question 8. [6 marks: 1, 1, 2, 2]

Maths department at Rossmoyne Senior High School purchased a new photocopier for \$3 250. Its value depreciates at a rate of 4.5 cents per copy. Let V_n be the value of the photocopier in dollars after n copies have been made, where $V_n = a + bn$.

- (a) State the value of the constant a and the value of the constant b .

$$a=3250 \quad b=-0.045$$

- (b) Calculate the value of the photocopier after 44 000 copies have been made.

$$V_{44000} = 3250 - 0,045 \times 44000 = \$1270$$

- (c) Determine n when $V_n = 2647$.

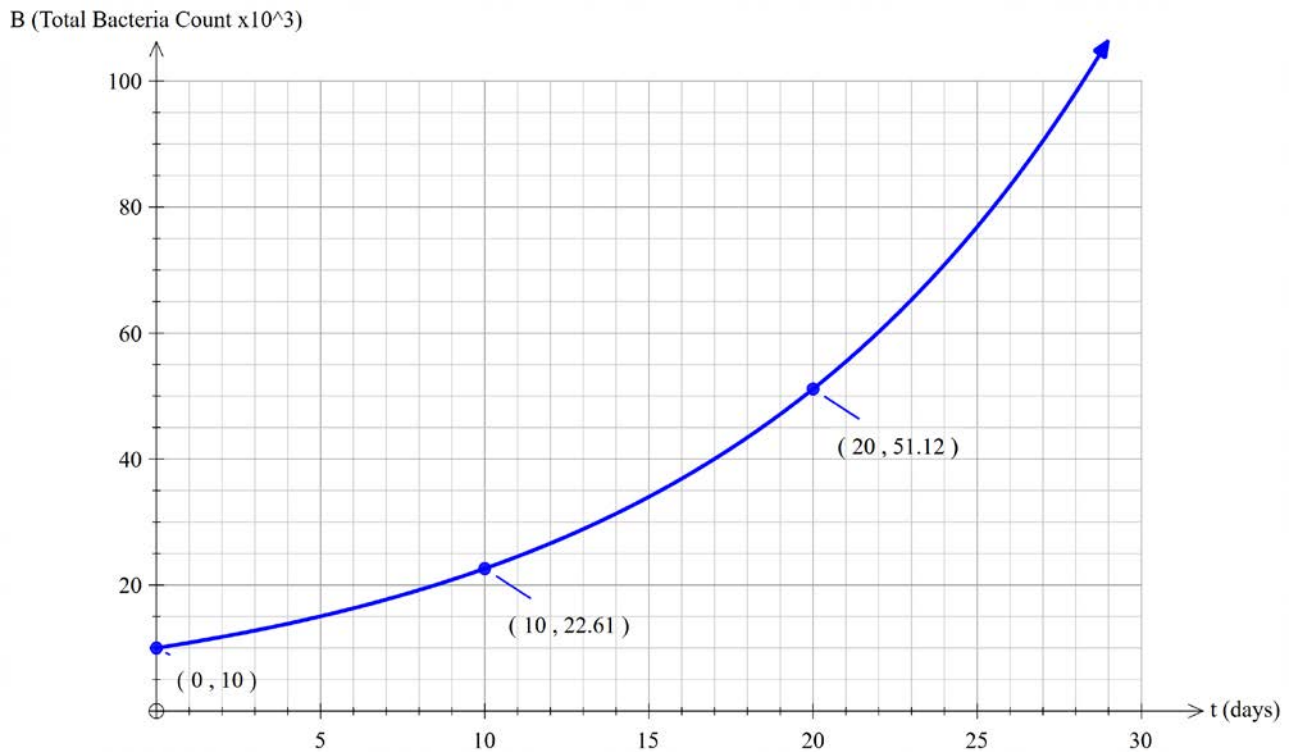
$$2647 = 3250 - 0.045 n$$
$$n=13400$$

- (d) The photocopier will be replaced as soon as its value falls below \$350. Determine the number of copies the photocopier will make before it is replaced.

$$350 = 3250 - 0.045 n$$
$$n=64444.4444444$$
$$n = 64445$$

Question 9. [7 marks: 2, 1, 2, 2]

Water collected from Serpentine River contained microorganisms. In the laboratory, a chemical cocktail was added to the water to aid in the proliferation of the microorganisms. The total bacteria (B) in the sample was measured in thousands for 28 days and was found to relate to time, t (days) by the formula, $B = a(k^t)$. The accompanying graph contains data points from the data logger.



- (a) Use an appropriate method to find the values of a and k .
Give the value of a to the nearest 100 and the values of k to 3 decimal places.

Use $(0, 10)$ $(10, 22.61)$ to solve a and k
 $a = 10$, $k = 1.085$

- (b) Predict the population after 40 days.

$$B = 10 \times 1.085^{40} = 261.33$$

- (c) How many days will it take for the total bacteria count to reach 150 000?

$$150 = 10 \times 1.085^t$$
$$t = 33.195 \text{ days or round to 34 days}$$

- (d) Water sample from the Swan River was mixed with the same chemical cocktail and studied over the same time period and its total bacteria count, C , is given as $C = 80(0.95)^t$. Determine when the two total bacteria counts are equal?

$$80 \times 0.95^t = 10 \times 1.085^t$$
$$t = 15.65 \text{ days}$$

Question 10. [7 marks: 1, 2, 2, 2]

The events A and B are such that $P(A) = 0.4$ and $P(B) = 0.5$.
Determine $P(A \cup B)$ in each of the following cases:

(a) A and B are mutually exclusive.

$$P(A \cup B) = 0.4 + 0.5 = 0.9$$

(b) $P(A \cup B') = 0.75$.

$$P(B \cap \bar{A}) = 1 - 0.75 = 0.25$$
$$P(A \cup B) = 0.4 + 0.25 = 0.65 = \frac{13}{20}$$

(c) A and B are independent.

$$P(A \cap B) = 0.4 \times 0.5 = 0.2$$
$$P(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$$

(d) $P(B | A') = \frac{7}{12}$

$$P(B|\bar{A}) = P(B \cap \bar{A})/P(\bar{A})$$
$$P(B \cap \bar{A}) = (1 - 0.4) \times \frac{7}{12} = 0.35$$
$$P(A \cup B) = 0.4 + 0.35 = 0.75$$